Course Plan

A. Course Handout

Institute/School Name	Chitkara University Institute of Engineering & Technology		
Department Name	Department of Computer Science and Engineering		
Programme Name	B.E (Computer Science & Engineering)		
Course Name	Advanced Data Structures	Session	Jan – June 2026
Course Code	22CS036	Semester/Batch	6th /2023
L-T-P(Per Week)	2-0-4	Course Credits	03
Pre-requisite	Basic knowledge of Programming Concepts	NHEQF Level	05[1]
Course Coordinator	Dr. Shivani Wadhwa	SDG Number	4,8,9[4]

1. Objectives of the Course

The scope of the course is to provide a comprehensive foundation for understanding core concepts of data structures and algorithms, and to develop strong problem-solving and analytical skills. Students will gain in-depth knowledge of arrays, strings, recursion, and backtracking, along with a solid grounding in object-oriented programming principles. The course emphasizes both linear and non-linear data structures and explores various algorithm design strategies through practical examples, enabling learners to analyze and evaluate data structures effectively for real-world applications. The objectives of the course are:

- to understand the detailed view of Arrays, Strings, Recursion, Backtracking.
- to build strong object-oriented programming fundamentals and proficiency in linear and non-linear data structures.
- to explore and implement various algorithm design strategies using illustrative examples.
- to analyze and evaluate different data structures for efficient problem-solving.
- to implement the concepts of data structures and algorithms by solving complex engineering problems and preparing well for interviews, competitions, and hackathons.

2. Course Learning Outcomes

After completion of the course, the student should be able to:

	Course Learning Outcome	*POs	NHEQF Level Descriptor[1]	Sessions
CLO01	Understand the detailed view of data structures and algorithms with underlying mathematics behind it.	PO1, PO2, PO4	Q1, Q2, Q6	11
CLO02	Revisit Object Oriented fundamentals along with the concepts of other linear data structures like Linked lists and Stacks and non-linear data structures like Graphs, Tries, Binary Trees and its variations; with main emphasis on Interview based questions.	PO1, PO8, PO9	Q3, Q5	19
CLO03	Explore various algorithm strategies such as DP, Greedy Method, Backtracking and Bit-masking.	PO2, PO3, PO5	Q2, Q4, Q6	28
CLO04	Analyse and evaluate different data structures and will be able to prepare well for Interview panels through numerical understanding of the concepts.	PO2, PO4	Q2, Q3, Q4, Q6	12
CLO05	Implement the concepts of data structures and algorithms on several forums like code-chef, coding ninjas, GFG and Hacker Rank.	PO7, PO8, PO10, PO11	Q6	20
Total Contact Hours				**90**

Revised Bloom's Taxonomy Terminology

* PO's available at (shorturl.at/cryzF)

**Cognitive Level =CL

***Knowledge Categories = KC

CLO-PO Mapping

CLO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	Type of Assessment's[3]
CLO01	H	M		L								L	Formative, Summative
CLO02	M							L	L				Formative, Summative
CLO03		L	M		M							M	Formative, Summative
CLO04		H		M									Formative, Summative
CLO05							L	M		H		M	Formative, Summative

H=High, M=Medium, L=Low

3. Recommended Books:

Textbooks:

B01: Introduction to Algorithms, Second Edition, Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, MIT Press, Cambridge, Massachusetts London, England McGraw-Hill Book Company, ISBN 0-262-03293-7.

B02: Computer Algorithms by E. Horowitz, S. Sahni and S. Rajsekran, Computer Science Press, New York, ISBN – 0-7167-8316-9.

B03: Data Structures and Algorithms in Java, Robert Lafore, Sams Publishing, 2nd edition, 2002.

B04: Data structures and algorithms in Java. John, Goodrich MT, Tamassia R, Goldwasser MH, wiley 2014

Reference Books:

B05: Data Structures with C (Schaum's Outline Series) (English, Paperback, Lipschutz Seymour), McGraw Hill Education India, ISBN: 9780070701984, 9780070701984

E-Resources:

https://library.chitkara.edu.in/subscribed-books.php

4. Other readings and relevant websites:

Serial No	Link of Journals, Magazines, websites and Research Papers
1.	https://onlinecourses.nptel.ac.in/noc22_cs26/preview
2.	https://www.youtube.com/watch?v=zWg7U0OEAoE
3.	https://iq.opengenus.org/list-of-advanced-data-structures/
4.	https://in.coursera.org/learn/advanced-data-structures
5.	https://www.geeksforgeeks.org/advanced-data-structures/
6.	https://www.youtube.com/@JennyslecturesCSIT/videos

[1] National Higher Education Qualification Framework Level, Refer to annexure

[1] NHEQF Level Descriptor, Refer to Annexure & Learning outcomes descriptors for qualification for all levels on the NHEQF

[3]Types of Assessments can be referred from Type of Assessments. Refer to Annexure.

[4]For SDG Mapping with Courses, Pl refer SDG Mapping policy for Courses

5. Recommended Tools and Platforms

- Visual Studio IDE
- Tespad

6. Course Plan:

Lecture Number	Topics	Textbook
1	**Detail Discussion of Course Handout (CHO)**	B05-Chpater-4
2-4	Arrays – 1: Advance Problems on Sliding Windows and Two pointers will be discussed.	B05-Chapter-4
5-6	Arrays – 2: Discussion Frequency Arrays and Prefix Arrays and Its Problems.	B05-Chapter-4
7-9	Binary Search: Binary Search, Its Implementation and Advance Binary Search problems will be discussed.	B05-Chapter-4
10-11	Array Advanced Algorithms – 1: Kadane's Algorithm, Prime Seive, Vectors and Amalgamation of the above topics will be discussed.	B05-Chapter-4
12-13	Strings: Implementation Problems on Strings, Two pointers in Strings and Problem on Frequency Arrays with respect to Strings.	B01-Chapter-32
14-15	Sliding Window – 1: Advance Problems on Sliding Window and Discussion of Deque STL.	B05-Chpater-4
16-17	Advanced String Algorithms: String Matching Algorithms like KMP, Z Function, and Rabin Karp will be discussed and problems based on that.	B01-Chapter-32
18-19	Array Interview Preparation class: Recently asked company problems based upon arrays, string and the above mentioned algorithms.	B01-Chapter-32

20-21	Recursion – 1: Basics of Recursion, basic problems on recursion to have better understanding of the call stack, Types of Recursion.	B05-Chapter-6
22-25	Recursion Backtracking – 1: Implementation based Problems on recursion will be discussed in a thorough manner with call stacks, Introduction to Backtracking and basic problem on backtracking to get the flow of it. Recursion Backtracking – 2: Advance problems on backtracking.	B05-Chapter-6
26-30	Recursion on Matrix: Discussion on how recursion and backtracking are used in general to solve problems on matrix/grid.	B05-Chapter-6
31-32	OOPS-1: Discussion on OOPs with real world example why it is important, Discussions on classes, objects and self invoked functions like Constructors, Copy Constructor, Copy Assignment Operators, Destructors.	B05-Chapter-1
33-35	OOPS-2: Discussion on characterises of OOPs like Encapsulation, Abstraction, Polymorphism, Inheritance, Dynamic Binding, Message Passing etc.	B05-Chapter-1
36-38	Linked List – 2: Interview Based Problems on LL.	B05-Chapter-5
ST-1 Syllabus (Lecture number 1-38)		
39-40	Stack Interview Questions: Interview Problems Based on Stacks.	B01-Chapter-10
41-42	Binary Trees – 1: Binary Trees (Definition, Implementation, Algorithms on Trees (DFS, BFS, etc.),Standard Problems on Trees)	B01-Chapter-12
43	Binary Trees – 2: Interview Problems Based on Trees.	B01-Chapter-12
44-45	Binary Search Trees – 1: Implementation of BST, Standard Problems on BST.	B01-Chapter-12
46-47	Binary Search Trees – 2: Interview Problems Based on Trees.	B01-Chapter-12
48-49	Hash maps – 2: Advance Hashing Techniques, Interview Problems based Hashing etc.	B01-Chapter-11
50-51	Priority Queue – 2: Advance Problems on Heaps.	B05-Chapter-6
52-53	DP – 1: DP, Types of DP, Importance of DP, 1-D DP company oriented questions	B01-Chapter-15
54-55	DP – 2: Discussion on 2D DP and Grid DP and Advance Problems based on that.	B01-Chapter-15
56-57	DP – 3: Knapsack and Selecting Distinct based DP.	B01-Chapter-15
58-59	DP – 4: Problems based on DP Patterns.	B01-Chapter-15
60-61	DP – 5: Multi-dimensional DP	B01-Chapter-15
62-63	DP – 6: DP on Trees.	B01-Chapter-15
64-67	Greedy Algorithms – 2: Advance Problems on Greedy.	B01-Chapter-16
68-69	Bit-masking – 2: Interview Based Problems on Bit Manipulations.	B01-Chapter-16
70-71	Number Theory: Prime Sieve, Segmented Sieve, Euclid Algorithm, Extended Euclid Algorithm and Its Application in solving linear Diophantine equations and multiplicative modulo inverse, Totient Equation, Fermat Little Theoram etc.	B01-Chapter-31
72-73	Maths: Big Integers, Combinatorics, Solving Linear Recurrences, Mathematical Expectation etc.	B01-Chapter-31
ST-2 Syllabus (Lecture number 39-73)		
74-77	Graphs-1: Undirected Graphs, Cycle Detection, shortest cycle in an undirected graphs etc, Directed Graphs, Topological sort, Cycle Detection in an directed graphs)	B01-Chapter-22
78-79	Graphs-2: Kosaraju algorithm, DSU and Problems based on the above mentioned topics.	B01-Chapter-22
80-81	Graphs-3: Minimum Spanning Trees, Kruskal's and Prism's Algorithm, Introduction to Weighted Graphs	B01-Chapter-23
82-83	Graphs-4: Dijkstras's, Bellman ford algorithms and Problems based on the above mentioned Topics.	B01-Chapter-24
84-85	Graphs-5: Advance problems on Graphs.	B01-Chapter-25, B01-Chapter-26
86-88	Tries: Introduction, Range Queries and Interview Problems on Tries.	B01-Chapter-21
89-90	Revision and Doubt Clearing Sessions	B01-Chapter-35
ETE Syllabus (Lecture number 1-90)		

7. **Delivery/Instructional Resources**

Lecture Number	Topics	CLO	Book No, CH No, Page No	TLM[1]	ALM[2]	Web References	Audio-Video
1	**Detail Discussion of Course Handout (CHO)**	CLO01	B05-Chpater-4	Lecture Discussion			
2-4	Arrays – 1: Advance Problems on Sliding Windows and Two pointers will be discussed.	CLO01	B05-Chapter-4	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/sliding-window-problems-identify-solve-and-interview-questions/	https://www.techiedelight.com/sliding-window-problems/
5-6	Arrays – 2: Discussion Frequency Arrays and Prefix Arrays and Its Problems.	CLO01	B05-Chapter-4	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/count-of-possible-arrays-from-prefix-sum-and-suffix-sum-arrays/	https://www.youtube.com/watch?v=tUmhU9RtGHA
7-9	Binary Search: Binary Search, Its Implementation and Advance Binary Search problems will be discussed.	CLO01	B05-Chapter-4	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/binary-search/	https://www.youtube.com/watch?v=YJeoQBevNVo
10-11	Array Advanced Algorithms – 1: Kadane's Algorithm, Prime Seive, Vectors and Amalgamation of the above topics will be discussed.	CLO01	B05-Chapter-4	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/largest-sum-contiguous-subarray/ https://algomaster.io/learn/dsa/kadane-s-algorithm-introduction	https://www.youtube.com/watch?v=oL9jzKm6uZQ
12-13	Strings: Implementation Problems on Strings, Two pointers in Strings and Problem on Frequency Arrays with respect to Strings.	CLO02	B01-Chapter-32	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/two-pointers-technique/	https://www.youtube.com/watch?v=9mb8w5Oomaw
14-15	Sliding Window – 1: Advance Problems on Sliding Window and Discussion of Deque STL.	CLO02	B05-Chpater-4	Lecture Discussion	Quiz, Challenge, Assignment	https://www.scribd.com/document/923761384/sliding-window-questions	https://www.youtube.com/watch?v=hnEQ0EmajJM
16-17	Advanced String Algorithms: String Matching Algorithms like KMP, Z Function, and Rabin Karp will be discussed and problems based on that.	CLO02 CLO03	B01-Chapter-32	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/rabin-karp-algorithm-for-pattern-searching/ https://medium.com/tech-in-	https://www.youtube.com/watch?v=BHG4auxBHZ0

						depth/string-matching-algorithms-271d50a2a265	
18-19	Array Interview Preparation class: Recently asked company problems based upon arrays, string and the above mentioned algorithms.	CLO02	B01-Chapter-32	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-50-array-coding-problems-for-interviews/	https://www.youtube.com/watch?v=_71ENYohRiE
20-21	Recursion – 1: Basics of Recursion, basic problems on recursion to have better understanding of the call stack, Types of Recursion.	CLO02	B05-Chapter-6	Lecture Discussion	Quiz, Challenge, Assignment	https://wankhedeshubham.medium.com/recursion-basic-part-1-6908302b8606	https://www.youtube.com/watch?v=IJDJ0kBx2LM
22-25	Recursion Backtracking – 1: Implementation based Problems on recursion will be discussed in a thorough manner with call stacks, Introduction to Backtracking and basic problem on backtracking to get the flow of it. Recursion Backtracking – 2: Advance problems on backtracking.	CLO02	B05-Chapter-6	Lecture Discussion	Quiz, Challenge, Assignment	https://www.hackerearth.com/practice/basic-programming/recursion/recursion-and-backtracking/tutorial/	https://www.youtube.com/watch?v=K6M4AsLZ-yg
26-30	Recursion on Matrix: Discussion on how recursion and backtracking are used in general to solve problems on matrix/grid.	CLO02	B05-Chapter-6	Lecture Discussion	Quiz, Challenge, Assignment	https://unstop.com/blog/oops-concept-in-cpp	https://www.youtube.com/watch?v=Ocu3T3yXoQw
31-32	OOPS-1: Discussion on OOPs with real world example why it is important, Discussions on classes, objects and self invoked functions like Constructors, Copy Constructor, Copy Assignment Operators, Destructors.	CLO02	B05-Chapter-1	Lecture Discussion	Quiz, Challenge, Assignment	https://unstop.com/blog/oops-concept-in-cpp	https://www.youtube.com/watch?v=oRQOiyO-kHg
33-35	OOPS-2: Discussion on characterises of OOPs like Encapsulation, Abstraction, Polymorphism, Inheritance, Dynamic Binding, Message Passing etc.	CLO02	B05-Chapter-1	Lecture Discussion	Quiz, Challenge, Assignment	https://beginnersbook.com/2013/03/oops-in-java-encapsulation-inheritance-polymorphism-abstraction/prot.html	https://www.youtube.com/watch?v=46T2wD3IuhM

36-38	Linked List – 2: Interview Based Problems on LL.	CLO02	B05-Chapter-5	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-50-linked-list-interview-question/	https://www.youtube.com/watch?v=FrGKsNBaq_4
39-40	Stack Interview Questions: Interview Problems Based on Stacks.	CLO03	B01-Chapter-10	Lecture Discussion	Quiz, Challenge, Assignment	https://www.youtube.com/watch?v=S9LUYztYLu4	https://www.geeksforgeeks.org/dsa/top-50-problems-on-stack-data-structure-asked-in-interviews/
41-42	Binary Trees – 1: Binary Trees (Definition, Implementation, Algorithms on Trees (DFS, BFS, etc.),Standard Problems on Trees)	CLO03	B01-Chapter-12	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/binary-tree-data-structure/	https://www.youtube.com/watch?v=4s1Tcvm00pA
43	Binary Trees – 2: Interview Problems Based on Trees.	CLO03	B01-Chapter-12	Lecture Discussion	Quiz, Challenge, Assignment	https://www.interviewcake.com/concept/java/binary-tree	https://www.youtube.com/watch?v=9D-vP-jcc-Y
44-45	Binary Search Trees – 1: Implementation of BST, Standard Problems on BST.	CLO03	B01-Chapter-12	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/binary-search-tree-data-structure/	https://www.youtube.com/watch?v=GrJueLa9IAI
46-47	Binary Search Trees – 2: Interview Problems Based on Trees.	CLO03	B01-Chapter-12	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-50-binary-search-tree-coding-problems-for-interviews/	https://www.youtube.com/watch?v=V97oYgN9cIE
48-49	Hash maps – 2: Advance Hashing Techniques, Interview Problems based Hashing etc.	CLO03	B01-Chapter-11	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-20-hashing-technique-based-interview-questions/	https://www.youtube.com/watch?v=xKCdp0jjZAw
50-51	Priority Queue – 2: Advance Problems on Heaps.	CLO03	B05-Chapter-6	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/priority-queue-using-binary-heap/	https://www.youtube.com/watch?v=HqPJF2L5h9U
52-53	DP – 1: DP, Types of DP, Importance of DP, 1-D DP company oriented questions	CLO04	B01-Chapter-15	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-20-dynamic-programming-interview-questions/	https://www.youtube.com/watch?v=HBON-ooIygo
54-55	DP – 2: Discussion on 2D DP and Grid DP and Advance Problems based on that.	CLO04	B01-Chapter-15	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/competitive-programming/dp-on-grids/	https://www.youtube.com/watch?v=RxAPGNpb6cc
56-57	DP – 3: Knapsack and Selecting Distinct based DP.	CLO04	B01-Chapter-15	Lecture Discussion	Quiz, Challenge, Assignment	https://cp-algorithms.com/dynamic_programming/	https://www.youtube.com/watch?v=4YaV8eUaM7Q

						knapsack.html	
58-59	DP – 4: Problems based on DP Patterns.	CLO04	B01-Chapter-15	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-50-dynamic-programming-coding-problems-for-interviews/	https://www.youtube.com/watch?v=qMfJegB20BY https://www.youtube.com/watch?v=sWnBVPRA0yQ
60-61	DP – 5: Multi-dimensional DP	CLO04	B01-Chapter-15	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/dp-problems-dimention-wise/	https://www.youtube.com/watch?v=tNWeB4jfmZA
62-63	DP – 6: DP on Trees.	CLO04	B01-Chapter-15	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/competitive-programming/dp-on-trees-for-competitive-programming/	https://www.youtube.com/watch?v=RuNAYVTn9qM
64-67	Greedy Algorithms – 2: Advance Problems on Greedy.	CLO04	B01-Chapter-16	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/greedy-algorithms/	https://www.youtube.com/watch?v=-wqKvpYGg1I
68-69	Bit-masking – 2: Interview Based Problems on Bit Manipulations.	CLO04	B01-Chapter-16	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-problems-on-bit-manipulation-for-interviews/	https://www.youtube.com/watch?v=qR0rygzxnE4
70-71	Number Theory: Prime Sieve, Segmented Sieve, Euclid Algorithm, Extended Euclid Algorithm and Its Application in solving linear Diophantine equations and multiplicative modulo inverse, Totient Equation, Fermat Little Theoram etc.	CLO04, CLO05	B01-Chapter-31	Lecture Discussion	Quiz, Challenge, Assignment	https://www.scribd.com/document/459497860/Seive-pdf	https://www.youtube.com/watch?v=KdePjukNs98
72-73	Maths: Big Integers, Combinatorics, Solving Linear Recurrences, Mathematical Expectation etc.	CLO04, CLO05	B01-Chapter-31	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/maths-for-data-structure-and-algorithms-dsa-a-complete-guide/	https://www.youtube.com/watch?v=AJkvIkNdTHg

74-77	Graphs-1: Undirected Graphs, Cycle Detection, shortest cycle in an undirected graphs etc, Directed Graphs, Topological sort, Cycle Detection in an directed graphs)	CLO04, CLO05	B01-Chapter-22	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/detect-cycle-undirected-graph/	https://www.youtube.com/watch?v=X2_tYUuthH8
78-79	Graphs-2: Kosaraju algorithm, DSU and Problems based on the above mentioned topics.	CLO04, CLO05	B01-Chapter-22	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/c/kosarajus-algorithm-in-c/	https://www.youtube.com/watch?v=R6uoSjZ2imo
80-81	Graphs-3: Minimum Spanning Trees, Kruskal's and Prism's Algorithm, Introduction to Weighted Graphs	CLO04, CLO05	B01-Chapter-23	Lecture Discussion	Quiz, Challenge, Assignment	https://sethuram52001.medium.com/minimum-spanning-tree-prims-and-kruskal-s-algorithm-134cb0a32212	https://www.youtube.com/watch?v=f30rxGvPWPo
82-83	Graphs-4: Dijkstras's, Bellman ford algorithms and Problems based on the above mentioned Topics.	CLO04, CLO05	B01-Chapter-24	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/bellman-ford-algorithm-dp-23/	https://www.youtube.com/watch?v=0vVofAhAYjc
84-85	Graphs-5: Advance problems on Graphs.	CLO04, CLO05	B01-Chapter-25, B01-Chapter-26	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-50-graph-coding-problems-for-interviews/	https://www.youtube.com/watch?v=LPNNM0-qWg4
86-88	Tries: Introduction, Range Queries and Interview Problems on Tries.	CLO04, CLO05	B01-Chapter-21	Lecture Discussion	Quiz, Challenge, Assignment	https://interviewing.io/tries-interview-questions	https://www.youtube.com/watch?v=MyiHeqtwOWQ
89-90	Revision and Doubt Clearing Sessions	CLO05	B01-Chapter-35	Lecture Discussion	Quiz, Challenge, Assignment	https://www.geeksforgeeks.org/dsa/top-100-data-structure-and-algorithms-dsa-interview-questions-topic-wise/	https://www.youtube.com/watch?v=sJhccRp__aI

8. Remedial Classes[4]

Slow Learners	Average Learners	Fast Learners
• Remedial Classes on Saturdays • Encouragement for improvement using Peer Tutoring • Use of Audio and Visual Materials • Use of Real-Life Examples	• Workshops • Formative Exercises used to highlight concepts and notions • E-notes and E-exercises to read ahead of the pedagogic material.	• Engaging students to hold hands of slow learners by creating a Peer Tutoring Group • Design solutions for complex problems • Design solutions for complex problems • Presentation on topics beyond those covered in CHO

9. Self-Learning[2]

Assignments to promote self-learning, a survey of contents from multiple sources.

S. No.	Topics	CLO	ALM	References/MOOCS
1	Introduction to OOPS	CLO01	Group Discussion	https://www.udemy.com/course/learning-data-structures-and-algorithms/ https://www.coursera.org/specializations/data-structures-algorithms
2	Advanced techniques of solving data models	CLO02 CLO03	Group Discussion	https://hackr.io/tutorials/learn-data-structures-algorithms https://www.edx.org/course/algorithms-and-data-structures
3	Recursion and Backtracking	CLO03 CLO04	Group Discussion	https://www.codingninjas.com/courses/onlline-c-plus-plus-course https://swayam.gov.in/nd1_noc20_cs85/preview
4	Dynamic Programming	CLO03 CLO04	Group Discussion	https://practice.geeksforgeeks.org/courses/dsa-self-paced
5	Graphs	CLO05	Group Discussion	https://nptel.ac.in/courses/106/102/106102064/

10. Delivery Details of Content Beyond Syllabus[3]

S. No.	Advanced Topics, Additional Reading, Research papers and any	CLO	POs	ALM	References/MOOCS
1	Advanced Algorithm Techniques	CLO01	PO1, PO9	Group Discussion	https://swayam.gov.in/nd2_cec19_cs04/preview
2	Interview Questions on Advance Data Structures	CLO02	PO2	Group Discussion	https://www.youtube.com/channel/UCu4ztYtW-Bg1KIfcLAULtVQ
3	Hashing	CLO03	PO10	Group Discussion	https://swayam.gov.in/nd1_noc20_cs85/preview

11. Evaluation Scheme & Components:

Assessment Type[4]	Evaluation Component[5]	Type of Component[6]	No. of Assessments[7]	% Weightage of Component	Max. Marks	Mode of Assessment	CLO
Summative	Component 2	Sessional Tests(STs)	02*	40%	40	Computer based online in Campus	1 – 2 1 – 4
Summative	Component 3	End Term Examination	01**	60%	60	Computer based online in Campus	1 - 5
Total				100%	100		

*Students will have to appear in all Sessional Tests.
**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

12. Syllabus of the Course:

Subject: Advance Data Structures / 23CSE….			
S. No.	**Topic (s)**	**No. of Sessions**	**Weightage %**
1	Arrays – 1, Arrays – 2, Array Advanced Algorithms-1 , Array Advanced Algorithms-2, Strings, Sliding Window – 1, Advanced String Algorithms, Array Interview Preparation class, Recursion Backtracking – 1, Recursion Backtracking – 2, Recursion on Matrix, OOPS-1, Linked List – 2	**38**	**42%**

[2] Refer to Annexure
[3] Refer to Annexure
[4] Refer to Annexure 2 of NCrF
[5] Refer to Annexure
[6] Refer to Annexure
[7] Refer to Annexure

ST-1 (Covering 42% syllabus)			
2	Stack Interview Questions, Binary Trees – 1, Binary Trees – 2, Binary Search Trees – 1, Binary Search Trees – 2, Hashmaps – 2, Priority Queue – 2, DP, Greedy Algorithms – 2, Bitmasking – 2,	**35**	**42%**
ST-2 (Covering 93% syllabus)			
3	Graphs, Advance problems on Graphs , Tries	**17**	**16%**
End Term 100% syllabus			

13. Academic Integrity Policy:

Education at Chitkara University builds on the principle that excellence requires freedom where Honesty and integrity are its prerequisites. Academic honesty in the advancement of knowledge requires that all students and Faculty respect the integrity of one another's work and recognize the importance of acknowledging and safeguarding intellectual property. Any breach of the same will be tantamount to severe academic penalties.

This Document is approved by:

Designation	**Name**	**Signature**
Course Coordinator	Dr. Shivani Wadhwa	
Head-Academic Delivery	Dr. Susheela Hooda	
Dean	Dr. Rupali Gill	
Date	31.01.2026	